



DANILO LEOCATA

IT Consultant · Backend & Data Science Engineer

CONTACT

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ABOUT ME

M.Sc. in Data Science with 6+ years of hands-on industry experience spanning backend engineering, cloud infrastructure, and software best practices. Academic background in machine learning and data-driven systems, complemented by dedicated coursework in software engineering, cloud architecture, and backend development patterns. Currently working as IT Consultant in fin-tech — leading projects end-to-end, driving modernization initiatives, and mentoring development teams.

EDUCATION

M.Sc. – Data Science

Oct. 2020 – Dec. 2023

🎓 **University of Catania, Italy**

Graduated **110/110**. Specialized in machine learning and data science, complemented by coursework in DevOps methodologies and software engineering. Thesis: *Development of Robotic Navigation Systems in Dynamic Environments based on Deep Reinforcement Learning* — resulted in a peer-reviewed publication at ICAART 2025 (see Publications).

Erasmus+

Sep. 2018 – Jun. 2019

✈ **Technical University of Sofia, Bulgaria**

All exams passed with full marks in English. International exposure across multiple cultures and languages.

B.Sc. – Computer Science

Oct. 2015 – Sep. 2019

🎓 **University of Catania, Italy**

Graduated **100/110**. Thesis: *Electronic Voting Procedures and Safeties*. Core technologies: **C, C++, Java, MySQL, React Native, PHP, JavaScript, HTML, CSS**.

PUBLICATIONS

CrowdSim++: Unifying Crowd Navigation and Obstacle Avoidance

ICAART 2025 — Vol. 2, pp. 533–542 DOI: 10.5220/0013192300003890

M. Rosano, **D. Leocata**, A. Furnari, G.M. Farinella

Proposes a unified RL-based simulator extending CrowdSim to jointly handle crowd navigation and obstacle avoidance in complex environments. Robots equipped with 2D LiDAR and a person sensor learn navigation policies that outperform single-task baselines.

WORK EXPERIENCE

IT Consultant – Full time

Mar. 2024 – present

 **Reply Technology Financial Services, Milano**

Fintech consultant for major banking clients across backoffice, trading, and customer-facing platforms. Fully autonomous in planning and delivering projects end-to-end — from requirements analysis and architectural design to implementation and production release. Led legacy modernization and microservices migration initiatives. Drove performance optimization through load testing and monitoring. Comfortable working independently as well as coordinating with colleagues and managing team activities, including the full delivery lifecycle: release management, testing pipelines, and enablement ticketing. Currently exploring and applying AI agents to streamline and accelerate daily development workflows.

Java, Spring Boot, Maven, Oracle DB, Kibana, RedHat, JPA, JDBC

Backend Developer – Full time

Jul. 2022 – Mar. 2024

 **Mongibello Tech, Catania**

Owned all microservices for game management on national and international gaming platforms. Operated autonomously across multiple concurrent projects, coordinating with international teams and delivering within tight deadlines.

Spring Boot, Gradle, Java, Groovy, MySQL, Bash

FullStack Developer – Part-time

Sep. 2021 – Jul. 2022

 **Synapses, Catania**

Bug fixing and feature development on the proprietary product; built an internal estimation tool from scratch.

Flutter, Java Spring Boot, Maven, Docker, Kubernetes

FullStack Developer – Full time

Dec. 2019 – Jun. 2021

 **Bizmate, Catania**

Developed multiple products from scratch: restaurant reservation web-app, License Manager, and Account Manager. Started with front-end maintenance (**AngularJS, jQuery**), then took full ownership of new builds.

Svelte, NodeJS, Docker, MongoDB, Keycloak, Redis, Bootstrap

FullStack Developer – Training programme

Oct. 2019 – Dec. 2019

 **Adecco, Bologna**

Adecco partner training programme focused on enterprise Java development.

Java, MySQL, Spring Boot, JPA, JDBC

CERTIFICATIONS

Assessing Management

Corporate Finance Institute

PROJECTS

Selected projects — full list on GitHub.

CrowdSim Real

2023

</> [Python](#)

Deep RL simulator for LiDAR-based robot navigation in complex, crowded floor plans.

Gamble platform with microservices

2023

</> [Java](#) · [Spring Boot](#)

Microservices-based online football betting platform following distributed systems best practices.

Svelte e-commerce

2022

</> [Svelte](#) · [Express](#)

Full-stack e-commerce demo.

Triplet Trashbin classifier

2022

</> [Python](#) · [PyTorch](#)

Triplet Neural Network for image classification using PyTorch Lightning.

Biobanksprec

2022

</> [Java](#) · [Kubernetes](#) · [AWS](#)

Monolith-to-microservices migration with CI/CD (GitHub Actions), Keycloak auth, and AWS deploy.

Football match predictor

2021

</> [Python](#)

Poisson-based model predicting international football match outcomes.

Trash bin classifier

2021

</> [Python](#) · [PyTorch](#)

CNN classifier for bin fill-level (empty / half / full).

EXTRACURRICULAR

- Good skills in the use of **Office suite**
- Good understanding of the **Spanish** language
- Drone pilot license — OPEN category A1/A3 (ENAC certified)